EXACT EXTERNAL INDICATORS AND THEIR IMPACT

Macroeconomic Indicators (Core – Mandatory)

1.1 Inflation & Price Stability

- Consumer Price Index (CPI)
- Core Inflation (CPI excluding food & energy)
- Producer Price Index (PPI)
- Inflation Expectations (survey-based or implied)

Impact

- Demand suppression or acceleration
- Pricing elasticity
- Cost pass-through

1.2 Interest Rates & Monetary Policy

- Central Bank Policy Rate
- Treasury Bill / Bond Yields (short & long term)
- Interbank Lending Rate
- Mortgage Rates

Impact

- Consumer borrowing
- Business investment
- Big-ticket purchases

1.3 Exchange Rates (FX)

- Local Currency ↔ USD
- Local Currency ↔ EUR
- Trade-weighted exchange rate index

Impact

- Import costs
- Export competitiveness
- Inflation transmission

1.4 GDP & Economic Growth

- GDP Growth Rate (QoQ, YoY)
- GDP per Capita
- Sectoral GDP (Manufacturing, Services, Agriculture)

Impact

- Overall demand capacity
- Sector-specific sales cycles

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2. Labor & Income Indicators

These determine **consumer spending power**.

2.1 Employment Metrics

- Unemployment Rate
- Labor Force Participation Rate
- Job Creation / Job Loss Rates

2.2 Income Metrics

- Average Wage Growth
- Real Disposable Income
- Minimum Wage Changes

Impact

- Volume sales
- Product mix (premium vs basic)

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3. Energy & Commodity Indicators (High Impact)

Especially critical in emerging and import-dependent markets.

3.1 Energy Prices

- Crude Oil Price (Brent / WTI)
- Fuel Pump Prices (diesel, petrol)
- Electricity Tariffs
- Gas Prices

Impact

- Transportation costs
- Manufacturing costs
- Retail pricing

3.2 Key Commodity Prices

(depend on industry)

- Food commodities (maize, wheat, rice)
- Metals (steel, copper, aluminum)
- Agricultural inputs (fertilizer)

Impact

- Cost of goods sold (COGS)
- Supply-side constraints

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4. Fiscal & Tax Policy Indicators (Critical for MacroSight™)

These are **non-linear shock drivers**.

4.1 Taxation

- VAT / Sales Tax Rate changes
- Excise Duty changes
- Import/Export Tariffs
- Corporate Tax Rate

Impact

- Immediate price shocks
- Demand contraction or substitution

4.2 Government Spending

- Budget Deficit / Surplus
- Public Infrastructure Spending
- Subsidy Programs

Impact

- Sector-specific demand spikes

- Cash circulation effects

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5. Financial System Indicators

These reflect **credit availability and stress**.

- Credit Growth Rate
- Consumer Loan Approval Rates
- Non-Performing Loan (NPL) Ratio
- Money Supply (M2)

Impact

- Ability to finance purchases
- Business expansion capacity

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6. Consumer Confidence & Sentiment (Leading Indicators)

These often **predict demand before sales move**.

- Consumer Confidence Index
- Business Confidence Index
- Purchasing Managers' Index (PMI)
- Retail Sales Index (macro-level)

Impact

- Forward-looking demand signals
- Inventory planning accuracy

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7. Trade & Supply Chain Indicators

Important for availability and pricing.

- Import Volume Index
- Export Volume Index
- Shipping Costs Index
- Port Throughput Volumes
- Global Supply Chain Pressure Index

Impact

- Stock availability

- Lead times
- Price volatility

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8. Demographic & Structural Indicators

Slow-moving but strategic.

- Population Growth Rate
- Urbanization Rate
- Household Size
- Age Distribution

Impact

- Long-term demand trends
- Product portfolio strategy

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9. Policy & Legislative Events (Qualitative → Quantified)

These must be encoded as **events**, not raw numbers.

- New tax laws
- Subsidy removals
- Import bans
- Price controls
- Regulatory compliance mandates

ML Representation

- Event flags
- Severity score
- Duration
- Affected sectors

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10. Industry-Specific Indicators (Configurable)

MacroSight™ should allow **industry plugins**. Examples:

- Telecom: data prices, penetration rates
- Retail: foot traffic indices
- Automotive: vehicle registration data

- Agriculture: rainfall, drought index
- Construction: building permits

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11. Indicator Classification for ML

Class	Examples
Continuous	CPI, FX, fuel prices
Categorical	Policy type, regulation
Event-based	Tax change, ban
Lagging	GDP
Leading	PMI, confidence

This allows **automatic feature learning** by ML models.

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12. Minimum Viable Indicator Set (Recommended)

For initial MacroSight™ deployment:

1. CPI / Inflation
2. Policy Interest Rate
3. Exchange Rate (USD)
4. Fuel Prices
5. VAT / Tax Changes
6. Consumer Confidence Index
7. GDP Growth
8. Unemployment Rate

This set alone delivers **strong forecasting lift**.

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13. Strategic Note (Important) Do **not** hardcode indicator importance.

Your ML models must:

- Learn indicator weights dynamically
- Adapt importance over time
- Detect regime changes

This aligns with your requirement:

"predict and learn without being explicitly programmed."

Api mapping of data sources

1. Core Macroeconomic and Development Indicators

a. World Bank Open Data API (Free)

Provides a comprehensive set of indicators including GDP, CPI, unemployment, population, trade, etc.

- **API Source:** World Bank Indicator API
- **Example endpoint:** https://api.worldbank.org/v2/indicator/NY.GDP.MKTP.CD?format=json (GDP)
- **Use for:**
  - GDP (current USD, growth, per capita) ○ Consumer Price Index (CPI) ○ Unemployment and labor statistics ○ Trade and demographic data
- **Notes:** Vast global coverage; returns JSON by default.

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b. Trading Economics API (Commercial with Free Tier)

Large repository of macroeconomic indicators by country. •

Primary API: TradingEconomics.com API

- **Use for:**
  - Consumer Price Index (CPI) ○ Inflation metrics ○ GDP growth and levels ○ Interest rates ○ Labor market variables ○ Retail sales, production indices
- **Notes:** Requires API key; good coverage and frequent updates.

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c. Financial Modeling Prep (FMP) Economics API

Good developer-oriented API with specific endpoints for economic indicators.

- **Example endpoints:**
  - /stable/economic-indicators?name=GDP ○ /stable/treasury-rates
- **Use for:**
  - GDP
  - Treasury yields / interest rate proxies ○ Economic calendar (release schedule) • **Free tier available with limits.

Provides OECD member country data via SDMX API.

- **Best for:**
  - CPI and core inflation ○ Unemployment rates ○ Interest rates ○ National accounts
- **Notes:** Free, SDMX-standard queries supported.

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e. IMF Data API (Free / SDMX)

International Monetary Fund provides IFS macroeconomic statistics.

- **Use for:**
  - Exchange rates ○ Balance of payments ○ Price indices ○ Money & banking statistics
- **Notes:** Uses SDMX; excellent global coverage and historical series.

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2. Specific Indicator APIs

a. Inflation API (Simple, Updated Monthly) •

Source: Inflation API (API-Ninjas)

- **Endpoints:**
  - /v1/inflation – current inflation ○ /v1/inflationhistorical – historical series
- **Use for:** Trend modeling, price pressure signals.

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b. Currency & FX Rates

- **Alpha Vantage FX endpoints** – FX pairs with daily/weekly historical data ○ Use for real exchange rate series
- **U.S. Treasury Fiscal Data API** – official exchange rate tables ○ Particularly for multi-currency rates in a standardized format

Note that free FX APIs (e.g., TwelveData) also exist but have usage limits — incorporate according to your throughput needs.

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3. Supplementary Indicator APIs / Sources

a. Commodities & Energy Prices

Many macro forecasting use cases require crude oil and key commodity price time series. Consider:

- **World Economic Indicators API (FinanceFlowAPI)** – CPI, employment, FX, crude prices •
- **EODHD Macro Indicators API** – inflation, GDP, unemployment, and other price indices

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b. Structural & Trade Policy Data

These are harder to find in a clean free API, but options include:

- **UN Comtrade API** – trade volumes and tariff data
- **FTA Tariff API** (US Dept. of Commerce) – tariff schedules for certain countries and goods
- **National Open Data Portals:** Many governments (e.g., data.gov, data.gov.uk, open.data.gov.sa) have macro datasets that can be programmatically ingested when available

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4. Consumer & Business Sentiment Indicators

These are often only available via paid or specialized services:

- **Trading Economics** – consumer confidence indices
- **FMP economic calendar / sentiment releases**
- **Central Bank & National Statistical Office feeds** (where available via open API)

Note: Consumer confidence is typically reported by private research bodies (e.g., The Conference Board) and requires licensing for API access.

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5. Integration Notes for MacroSight™ Backend

a. Source Prioritization

Build your indicator ingestion layer in Django such that each source adapter:

- **Normalizes indicator names to your domain taxonomy**
- **Handles rate limits and pagination**
- **Stores timestamps and country identifiers consistently**

Example adapter pattern: class WorldBankAdapter:

```
def fetch_gdp(country_code, start_date, end_date):
    ...
def fetch_inflation(...):
    ...
```

b. Frequency

Match API update cadence to your forecast horizon:

- Monthly updates for CPI, GDP
- Daily or weekly for FX and commodity prices
- Publish economic calendar events and flags

c. Governance

Store metadata such as:

- Source API name
- Retrieval timestamp
- Raw payload hash
- Version of the indicator series for auditability and reproducibility.